

A The Aglet Merchant

Time limit: 1s

Unfortunately, you work for an aglet merchant who has a vast fleet of caravans to transport his wares across a desert. Recently, a robber has been plundering his caravans. As his lowly servant, he commands you to redesign the distribution process to ensure that he stays filthy rich. As the slimy, greedy pessimist he is, the merchant wants you to maximize how much he is guaranteed to earn even if the robber attacks the worst possible caravan, and steals all the aglets from it. Since the robber cannot be everywhere at once, only one caravan can be attacked. His caravans travel towards cities, and along the way, there can be intermediary stops, they call trading posts. Trading posts are just places where caravans can go to and deliver the wares to the next caravans, but unlike cities, the aglets never end there, they just pass through. Since the robber might not attack, and the merchant does not want to lose profit, he will always send as many aglets as he can, limited only by the caravan's capacity and the limits of what he is allowed to import into a city, set by the Trade Federation.



Picture of aglets

Input

The input consists of:

- One line containing two numbers C and P , where $1 \leq C \leq 100$ is the price to produce an aglet, and $C < P \leq 1000$ is the selling price of an aglet.
- A second line contains three numbers T , N and R , where $0 \leq T \leq 500$ is the number of trading posts that the caravans can pass through, $1 \leq N \leq 100$ is the number of cities he exports aglets to and $1 \leq R \leq 1500$ is the number of caravans that go between trading

posts and cities. Trading posts are numbered 1 to T, and the cities are numbered T+1 to T+N. The merchant ships all his aglets from his warehouse at 0.

- Then comes N lines, each containing an integer which is the amount of aglets The Trade Federation has allowed you to import into the N-th city.
- Lastly R lines, each containing 3 integers, representing a caravan. The first integer is the starting point, the second is the endpoint, both of which represent a trading post, a city or his warehouse. The third is the amount of aglets the caravan can carry, which is always lower than 2^{31} . There can only be one caravan between a given starting point and ending point.

Output

If the merchant can make money on the trade route no matter where the robber attacks, output two integers. The first one is the number of aglets that will still reach the city if the robber attacks the caravan with the most aglets. The second is the amount of profit the merchant will make in the same case, if all aglets reaching the city is sold.

If it is possible that he can not make a profit, output "Not worth".

Sample Input 1

1 2 3 1 6 6 0 1 10 0 2 9 0 3 8 1 4 10 2 4 9 3 4 8	4 2
---	-----

Sample Output 1

Sample Input 2

2 3 3 1 6 3 0 1 10 0 2 11 0 3 12 1 4 10 2 4 11 3 4 12	Not worth
---	-----------

Sample Output 2

Sample Input 3

```
1 4
3 2 7
10
20
0 1 5
0 2 7
1 3 6
2 3 5
3 4 1
3 5 8
2 5 4
```

Sample Output 3

```
5 8
```